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TREND-BASED MOCK PAPER

GATE Electronics and Communication Engineering

Mock Paper — 01

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electronics and Communication Engineering
Paper:	Mock 01
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

*Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.*

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Sum of first 100 natural numbers:

- (A) 1000
- (B) 5050
- (C) 4950
- (D) 5000

Q2. [1-MCQ] [1 mark]

If train 100m crosses pole in 10s, speed:

- (A) 10 m/s
- (B) 36 km/h
- (C) 18 km/h
- (D) 72 km/h

Q3. [1-MCQ] [1 mark]

Choose synonym of 'trivial':

- (A) Significant
- (B) Insignificant
- (C) Important
- (D) Major

Q4. [1-NAT] [1 mark]

Choose opposite of 'benevolent':

(Numerical Answer Type — Enter the numerical value)

Section B: Electronics and Communication Engineering

Q5. [2-MCQ] [2 marks]

Energy signal has:

- (A) Finite E, zero P
- (B) Infinite E
- (C) Finite P
- (D) Both finite

Q6. [2-NAT] [2 marks]

FT of $x(2t-1)$ given $X(j\omega)$:

(Numerical Answer Type — Enter the numerical value)

Q7. [2-NAT] [2 marks]

Auto-correlation FT =

(Numerical Answer Type — Enter the numerical value)

Q8. [2-MCQ] [2 marks]

Parseval relates:

- (A) Time and freq energy
- (B) Time only
- (C) Freq only
- (D) Power only

Q9. [2-MCQ] [2 marks]

FS of $f(t)=t$ on $(-\pi, \pi)$ has only:

- (A) Sine
- (B) Cosine
- (C) Both
- (D) DC only

Q10. [2-NAT] [2 marks]

FS of $f(t)=t$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q11. [2-MCQ] [2 marks]

Energy signal has:

- (A) Finite E, zero P
- (B) Infinite E
- (C) Finite P
- (D) Both finite

Q12. [2-NAT] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-MCQ] [1 mark]

$i^4 =$:

- (A) 0
- (B) 1

- (C) -1
- (D) i

Q14. [1-MCQ] [1 mark]

$P(A \cap B) = ?$

- (A) $P(A) + P(B)$
- (B) $P(A) + P(B) - P(A \cap B)$
- (C) $P(A)P(B)$
- (D) $P(A)/P(B)$

Q15. [1-NAT] [1 mark]

Integral of $1/x$ dx:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q17. [2-MCQ] [2 marks]

Fourier series of $f(x) = x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q18. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q20. [2-MCQ] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q21. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6

(D) 0.4

Q22. [2-MCQ] [2 marks]

Op-amp integrator $R=10k$, $C=1\mu F$. Unit step after 1ms:

- (A) -0.1V
- (B) -0.01V
- (C) -0.001V
- (D) -1V

Q23. [2-MCQ] [2 marks]

CMRR of diff amp depends on:

- (A) Only R_C
- (B) Only β
- (C) Active load matching
- (D) Temperature

Q24. [2-NAT] [2 marks]

Frequency response RC integrator (high-pass) corner:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

CE amp $h_{fe}=100$, $h_{ie}=1k$, $R_C=2k$. Voltage gain:

- (A) -200
- (B) -100
- (C) -150
- (D) -50

Q26. [2-NAT] [2 marks]

CC (emitter follower) has:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

Frequency response RC integrator (high-pass) corner:

- (A) $1/(2\pi RC)$
- (B) RC
- (C) $1/RC$
- (D) $2\pi RC$

Q28. [2-NAT] [2 marks]

CMRR of diff amp depends on:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-NAT] [2 marks]

Plane wave in $\epsilon_r=4$, $\mu_r=1$. Wavelength:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

$Z_0=50\Omega$, $Z_L=100\Omega$. Reflection coefficient:

- (A) 0.33
- (B) 0.5

- (C) 0.67
- (D) 1.0

Q31. [2-MSQ] [2 marks]

Poynting vector $S =$

- (A) $E \times D$
- (B) $E \times H$
- (C) $H \times E$
- (D) $B \times H$

Q32. [2-MCQ] [2 marks]

Skin depth in copper at 1MHz:

- (A) 66 microns
- (B) 66 mm
- (C) 6.6 microns
- (D) 6.6 mm

Q33. [2-MCQ] [2 marks]

Nyquist rate for $f_{\max}=4\text{kHz}$:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 16kHz

Q34. [2-MCQ] [2 marks]

NBFM bandwidth approx:

- (A) $2f_m$
- (B) $2(\Delta f + f_m)$
- (C) Δf
- (D) f_m only

Q35. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-NAT] [2 marks]

AM signal carrier 1kW, modulation 0.5. Total power:

(Numerical Answer Type — Enter the numerical value)

Q37. [1-MCQ] [1 mark]

Maxwell has how many equations:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q38. [1-NAT] [1 mark]

Characteristic impedance $Z_0 =$:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Skin depth increases with:

- (A) Frequency
- (B) Conductivity
- (C) Both
- (D) Neither

Q40. [2-MCQ] [2 marks]

Waveguide supports:

- (A) TEM
- (B) TE/TM
- (C) Both
- (D) None

Q41. [1-MCQ] [1 mark]

Common emitter has:

- (A) High input Z
- (B) Low input Z
- (C) Zero gain
- (D) Unity

Q42. [1-NAT] [1 mark]

Ideal op-amp open-loop gain:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-NAT] [1 mark]

FET is:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-NAT] [1 mark]

;

(Numerical Answer Type — Enter the numerical value)

Q45. [1-MCQ] [1 mark]

RC oscillator frequency:

- (A) $f = 1/(2\pi RC)$
- (B) $f = 1/RC$
- (C) $f = 1/(4\pi RC)$
- (D) $f = 2\pi RC$

Q46. [1-MCQ] [1 mark]

SCR has:

- (A) 2 terminals
- (B) 3
- (C) 4
- (D) 5

Q47. [1-MCQ] [1 mark]

n-type majority carriers:

- (A) Holes

- (B) Electrons
- (C) Both equal
- (D) None

Q48. [1-MCQ] [1 mark]

MOSFET is:

- (A) BJT
- (B) Unipolar
- (C) Bipolar
- (D) None

Q49. [1-MCQ] [1 mark]

Zener diode used for:

- (A) Rectification
- (B) Voltage regulation
- (C) Amplification
- (D) None

Q50. [1-MCQ] [1 mark]

ROM is:

- (A) Read only
- (B) Read write
- (C) Volatile
- (D) Cache

Q51. [1-MCQ] [1 mark]

D flip-flop follows input after:

- (A) Same instant
- (B) One clock
- (C) Two clocks
- (D) Never

Q52. [1-NAT] [1 mark]

4-bit binary counter counts:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

TTL stands for:

- (A) Transistor-Transistor Logic
- (B) Tunnel Transistor Logic
- (C) Totem Pole Logic
- (D) None

Q54. [1-MCQ] [1 mark]

Transfer function is defined:

- (A) In time domain
- (B) Laplace
- (C) Fourier
- (D) Z-domain

Q55. [1-MCQ] [1 mark]

Nyquist criterion uses:

- (A) Polar plot
- (B) Root locus
- (C) Routh
- (D) Bode

Q56. [1-MCQ] [1 mark]

Routh-Hurwitz determines:

- (A) Stability
- (B) Time response
- (C) Frequency
- (D) Sensitivity

Q57. [1-NAT] [1 mark]

Lead compensator:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-NAT] [1 mark]

ASK modulation varies:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-NAT] [1 mark]

FM has:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-NAT] [1 mark]

PCM stands for:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

AM stands for:

- (A) Amplitude Modulation
- (B) Angle Modulation
- (C) Analog Modulation
- (D) None

Q62. [2-MSQ] [2 marks]

Superposition theorem applies to:

- (A) Linear only
- (B) Non-linear
- (C) Both
- (D) Time-varying only

Q63. [2-MCQ] [2 marks]

Wheatstone bridge null condition:

- (A) $R_1/R_2 = R_3/R_4$
- (B) $R_1 \cdot R_4 = R_2 \cdot R_3$
- (C) Both A and B
- (D) $R_1 + R_2 = R_3 + R_4$

Q64. [2-MCQ] [2 marks]

4-bit Johnson counter states:

- (A) 4
- (B) 8
- (C) 16
- (D) 10

Q65. [2-MCQ] [2 marks]

$F(A,B,C,D) = \sum m(0,1,2,3,4,5,8,9,10)$. Minimal:

- (A) $A' + B'C$
 - (B) $A'D' + A'C'$
 - (C) $B' + A'D'$
 - (D) $A'B' + A'D'$
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1NAT	A	1
5	1MCQ	B	1
6	1MCQ	B	1
7	1NAT	B	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	B	1
13	1NAT	C	1

14	1NAT	B	1
15	1NAT	C	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1MCQ	B	1
19	1NAT	A	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1NAT	B	1
25	1NAT	A	1
26	1NAT	A	1
27	1NAT	A	1
28	1MCQ	A	1

29	1MCQ	C	1
30	1NAT	A	1
31	2MCQ	D	2
32	2MCQ	B	2
33	2NAT	A	2
34	2MCQ	B	2
35	2MCQ	A	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	C	2
40	2MCQ	A	2
41	2NAT	A	2
42	2NAT	A	2
43	2NAT	A	2

44	2MCQ	A	2
45	2MSQ	A	2
46	2MSQ	B	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2NAT	B	2
50	2MCQ	A	2
51	2NAT	A	2
52	2NAT	A	2
53	2NAT	C	2
54	2MCQ	A	2
55	2NAT	A	2
56	2MCQ	B	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2MCQ	B	2
62	2NAT	B	2
63	2MCQ	C	2
64	2MCQ	B	2
65	2NAT	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Complex Analysis | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: SCR | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Semiconductors | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: MOSFET | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Electronic Devices | Topic: Diodes | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: Amplifiers | Difficulty: moderate

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: Op-Amps | Difficulty: moderate

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Circuits | Topic: FET | Difficulty: moderate

Q15.

Answer: C

Solution:

Correct Answer: C

Subject: Analog Circuits | Topic: BJT | Difficulty: moderate

Q16.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Circuits | Topic: Oscillators | Difficulty: easy

Q17.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Memory | Difficulty: difficult

Q18.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Circuits | Topic: Flip-Flops | Difficulty: difficult

Q19.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Counters | Difficulty: difficult

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Circuits | Topic: Logic Families | Difficulty: moderate

Q21.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Transfer Function | Difficulty: difficult

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: easy

Q23.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Stability | Difficulty: moderate

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: difficult

Q25.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital | Difficulty: moderate

Q26.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Analog | Difficulty: difficult

Q27.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Digital Comm | Difficulty: moderate

Q28.

Answer: A

Solution:

Correct Answer: A

Subject: Communication Systems | Topic: Modulation | Difficulty: easy

Q29.

Answer: C

Solution:

Correct Answer: C

Subject: Electromagnetics | Topic: Maxwell Equations | Difficulty: moderate

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Electromagnetics | Topic: Transmission Line | Difficulty: difficult

Q31.

Answer: D

Solution:

Correct Answer: D

Subject: Electromagnetics | Topic: Wave Propagation | Difficulty: moderate

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Electromagnetics | Topic: Waveguides | Difficulty: easy

Q33.

Answer: A

Solution:

$$v = c/\sqrt{4 \times 1} = c/2. \lambda = v/f = \lambda_0/2.$$

Subject: EMFT | Topic: EMFT | Difficulty: easy

Q34.

Answer: B

Solution:

$$f_s \geq 2 \times 4\text{kHz} = 8\text{kHz}.$$

Subject: Communication | Topic: Communication | Difficulty: easy

Q35.

Answer: A

Solution:

$$\text{Gamma} = (100-50)/(100+50) = 50/150 = 1/3 = 0.33.$$

Subject: EMFT | Topic: EMFT | Difficulty: easy

Q36.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Energy: finite E, P=0. Power: finite P, E=infinity.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q38.

Answer: A

Solution:

$$V_{\text{out}} = -t/(RC) = -100 \times t. \text{ At } 1\text{ms: } -0.1\text{V}.$$

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: difficult

Q39.

Answer: C

Solution:

CMRR depends on active load matching. Active load improves CMRR.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q40.

Answer: A

Solution:

NBFM: BW approx $2 \times f_m$ (same as AM). WBFM: Carson rule.

Subject: Communication | Topic: Communication | Difficulty: easy

Q41.

Answer: A

Solution:

RC high-pass cutoff: $1/(2\pi RC)$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q42.

Answer: A

Solution:

$$x(2(t-0.5)): \text{scale } 1/2, \text{ shift } e^{(-j\omega/2)}.$$

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q43.

Answer: A

Solution:

Wiener-Khinchin: FT of $R_{xx}(\tau) = |X(j\omega)|^2$.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q44.

Answer: A

Solution:

$$A_v = -h_{fe} R_C / h_{ie} = -100 \cdot 2000 / 1000 = -200.$$

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q45.

Answer: A

Solution:

Superposition requires linearity (homogeneity + additivity). Does NOT apply to non-linear.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q46.

Answer: B

Solution:

$S = E \times H$. Direction: power flow. Magnitude: power per unit area.

Subject: EMFT | Topic: EMFT | Difficulty: difficult

Q47.

Answer: A

Solution:

Parseval: energy in time domain = energy in frequency domain.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q48.

Answer: A

Solution:

Odd function: only sine terms.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q49.

Answer: B

Solution:

CC: $A_v \sim 1$, high input Z, high current gain, low output Z.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: moderate

Q50.

Answer: A

Solution:

RC high-pass cutoff: $1/(2\pi RC)$.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q51.

Answer: A

Solution:

$$P_{\text{total}} = P_c(1 + m^2/2) = 1000 \cdot (1 + 0.125) = 1125\text{W} = 1.125\text{kW}.$$

Subject: Communication | Topic: Communication | Difficulty: moderate

Q52.

Answer: A

Solution:

Odd function: only sine terms.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q53.

Answer: C

Solution:

CMRR depends on active load matching. Active load improves CMRR.

Subject: Analog Electronics | Topic: Analog Electronics | Difficulty: easy

Q54.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q55.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

Energy: finite E, P=0. Power: finite P, E=infinity.

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q58.

Answer: B

Solution:

Johnson (4-bit): 8 states.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q59.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: A

Solution:

$\delta = \sqrt{2/(\omega \mu \sigma)}$. At 1MHz, δ approx 66 microns.

Subject: EMFT | Topic: EMFT | Difficulty: moderate

Q61.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: B

Solution:

$$f_s \geq 2f_{\max} = 8\text{kHz}.$$

Subject: Signal Systems | Topic: Signal Systems | Difficulty: moderate

Q63.

Answer: C

Solution:

Null: $R_1/R_2 = R_3/R_4$, which implies $R_1 \cdot R_4 = R_2 \cdot R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q64.

Answer: B

Solution:

K-map: $F = A'(D' + B'C') = A'D' + A'B'C'$. Closest: $A'D' + A'C'$.

Subject: Digital Electronics | Topic: Digital Electronics | Difficulty: moderate

Q65.

Answer: A

Solution:

$$P_{\text{total}} = P_c(1+m^2/2) = 1000(1+0.125) = 1125\text{W} = 1.125\text{kW}.$$

Subject: Communication | Topic: Communication | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Signal Systems: 8 questions, 16 marks
Engineering Mathematics: 9 questions, 15 marks
Analog Electronics: 7 questions, 14 marks
EMFT: 4 questions, 8 marks
Communication: 4 questions, 8 marks
Electromagnetics: 4 questions, 6 marks
Analog Circuits: 5 questions, 5 marks
General Aptitude: 4 questions, 4 marks
Electronic Devices: 4 questions, 4 marks
Digital Circuits: 4 questions, 4 marks
Control Systems: 4 questions, 4 marks
Communication Systems: 4 questions, 4 marks
Network Theory: 2 questions, 4 marks
Digital Electronics: 2 questions, 4 marks

Difficulty Distribution:

Easy: 18 | Moderate: 37 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.